LEVERAGE AND STRUCTURAL INFORMATION IN NOISY STOCHASTIC-VOLATILITY PRICES

J. VIDAL LLAURADÓ

Abstract. Leverage can preserve information about a hidden volatility drift as covariance shrinks, provided its response propagates through the latent dynamics. For a bivariate Ornstein–Uhlenbeck model observed through noisy log-price levels on a fixed grid, we prove exact identification at positive covariance and adaptive squared drift risk of order $\min\{1, [n(s^2 + se^2)]^{-1}\}$, where n counts observation blocks, s is the covariance scale and e the leverage amplitude. Matching lower bounds on specified drift neighborhoods use the full price law. Dynamically exciting leverage thus permits learning where zero leverage prevents uniform consistency. Nuisance parameters are unknown; confidence coverage includes zero covariance and absent excitation. Precision requires fixed excitation and residual covariance margins, fixed positive level noise, a nonaliased drift branch and a sufficiently small, unevaluated covariance ceiling.

1. When leverage changes learnability

When volatility fluctuations are small, noisy prices may contain little evidence about the dynamics generating them. The persistence and interaction of hidden volatility factors can then be difficult to estimate, even in a long sample. We use noisy price levels, with hidden volatility, and ask when return–volatility dependence makes those dynamics learnable. The answer depends on how the signed leverage response propagates through the latent system.

The sample grows through n observation blocks on a fixed grid. In the bivariate Ornstein–Uhlenbeck model studied here, the state covariance is $\Sigma = sV$, with $\operatorname{tr} V = 2$; hence $s = \operatorname{tr}\Sigma/2$, while the state diffusion amplitude is proportional to $\sqrt{s}$. Covariance responses have size s, whereas signed leverage responses have size $\sqrt{s}\, e$, where e is the Frobenius leverage amplitude. If the dynamics propagate that response across both state coordinates, leverage supplies information about every entry of the drift matrix.

At positive covariance and with a fixed excitation margin at the truth, an exact nonlinear moment inverse identifies the whole drift against alternatives throughout the admitted parameter class, with all nuisance parameters unknown. A single observable estimator adapts to the covariance and leverage amplitudes. Its squared drift risk is bounded by a constant times

$$\min\{1, [n(s^2 + se^2)]^{-1}\}.$$

Lower bounds for the full observed-price law match this rate on specified drift neighborhoods. No estimator based on those prices improves its worst-case order. For $s_n = n^{-a}$, $1/2 < a < 1$, covariance-only information ns_n^2 tends to zero, while $n(s_n^2 + s_n e^2)$ diverges for fixed nonzero dynamically exciting leverage. Under the neighborhood hypothesis of Theorem 3.1, this separates worst-case failure of uniform drift consistency from consistent estimation on the same noisy grid.

Confidence sets translate this information boundary into uncertainty about a fitted drift. Our sets have honest coverage over the full admitted class, including zero covariance and absent excitation. Their expected squared contrast diameter has the same upper order at excited positive-scale truths, with matching worst-case lower bounds on the specified neighborhoods. These precision guarantees require fixed excitation and residual covariance margins and a sufficiently small, unevaluated covariance ceiling.

2020 *Mathematics Subject Classification.* 62M05, 62F12, 62F25, 62P05.

Key words and phrases. Stochastic volatility, leverage, weak identification, noisy price observations, minimax estimation, honest confidence sets.

Signed leverage moments have long been used to study latent volatility. The correlated exponential-OU response and its diffusion-amplitude order occur in Masoliver and Perelló [6]. Ahsan, Dufour and Rodriguez-Rondon [3] study signed moments and recovery of unknown leverage and latent dynamics in a discrete scalar higher-order model with specified lag support and regular fixed-parameter theory. Here an exact integrated-price calculation allows unknown carry, covariance, loading and level noise; propagation determines which matrix drift directions benefit from the response.

Inference under weak identification is also established. Cheng, Renault and Sangrey [1] study volatility risk prices through leverage under reduced-form regularity assumptions supporting uniform inference. Ahsan and Dufour [2] develop identification-robust stochastic-volatility tests using high-frequency instruments and nuisance projection. Our confidence construction follows these principles, with diameter governed by the information about the hidden drift as covariance shrinks.

A moment inverse alone leaves open whether other features of the prices could give a better rate. Abraham, Naulet and Gassiat [8] connect weak moment geometry with full-sequence information for finite-state hidden models. Gloter [7] studies price-based drift inference, including noisy prices on a shrinking grid over a growing horizon. Chong, Hoffmann, Liu, Rosenbaum and Szymanski [9] prove actual-price minimax results for rough volatility in fixed-horizon infill with independent drivers and without microstructure noise. Our lower bounds concern the full noisy-price law on a fixed grid as covariance shrinks. Structural-shock identification with measurement error provides broader context in Gregoir and Meddahi [4].

2. The noisy-price experiment

Fix $N \geq 20$, $\Delta > h > 0$, $\delta = h/N$ and $r = \delta$. For $n \geq 2$ the observation times are all distinct elements of

$$\mathcal{T}_n = \{k\Delta + j\delta : 0 \leq k \leq n+1,\ 0 \leq j \leq N\}. \tag{1}$$

Both asset coordinates are observed:

$$P_i(t) = X_i(t) + \varepsilon_i(t), \qquad dX_i(t) = b_i\,dt + e^{Y_i(t)}dU_i(t), \quad i = 1, 2. \tag{2}$$

The clean initial log prices are fixed known constants. There is exactly one error per coordinate and physical time,

$$\varepsilon_i(t) \sim N(0, \nu_i),$$

independent across coordinates and distinct times and independent of all drivers. Reusing a time reuses its error. Here ν_i is a variance. Known bounds are

$$|b_i| \leq B_c, \qquad \mu_- \leq \mu_i \leq \mu_+, \qquad 0 < \nu_- \leq \nu_i \leq \nu_+ < \infty. \tag{3}$$

The carry b is an unknown constant log carry; there is no compensation term in (2). The experiment is the sigma-field of every noisy level in (1), including the initial noisy level and the observations adjoining the gaps. Grid spacings and noise bounds remain fixed as n grows and covariance shrinks.

The state is a two-sided stationary Gaussian OU process:

$$dY = M(Y - \mu)dt + A\,dB, \qquad \Sigma = sV, \qquad A = \sqrt{s}A_0, \qquad A_0 = \mathrm{chol}[-(MV + VM^\top)]. \tag{4}$$

Cholesky is lower triangular with positive diagonal. Its convention fixes the loading coordinates. The stationary prehistory has the model-implied law and is unobserved. At $s = 0$ set $A = 0$ and $Y = \mu$. The state coordinates are fixed by the outputs e^{Y_i}; there is no unknown output mixing.

The state driver B is standard two-dimensional Brownian motion, and

$$d\langle U\rangle = Q\,dt, \quad Q = \begin{pmatrix} 1 & q \\ q & 1 \end{pmatrix}, \qquad d\langle B, U\rangle = L\,dt. \tag{5}$$

Thus the row of L records a state-driver coordinate and the column a return-driver coordinate. Fix $0 < \kappa < 1$, $q_{\max} < 1$, and require

$$|q| \leq q_{\max}, \qquad Q - L^\top L \succeq \kappa I. \tag{6}$$

The full L, including support, rank, orientation and amplitude, is unknown. Taking traces gives $\|L\|_F^2 \le 2(1-\kappa)$. At $Q = I$, $\|L\|_{\mathrm{op}}^2 \le 1-\kappa$, so rank-one amplitudes can reach $\sqrt{1-\kappa}$, while full-rank Frobenius amplitudes can reach $\sqrt{2(1-\kappa)}$.

Fix a compact class $\mathcal{K}$ of (M, V) with

$$V = V^\top, \quad \operatorname{tr} V = 2, \quad v_- I \preceq V \preceq v_+ I, \qquad -(MV + VM^\top) \succeq \omega_- I, \tag{7}$$

where $v_-, \omega_- > 0$, on the prescribed branch

$$|\operatorname{Im} \lambda(M)| < \pi/\Delta. \tag{8}$$

Indeed $D = V^{-1/2} M V^{1/2}$ satisfies $D + D^\top \preceq -(\omega_-/v_+) I$. This proves strict stability and uniform exponential decay. Compactness gives a strict nonaliasing margin. Defective matrices are allowed. The Lyapunov equation in (4) gives stationary covariance sV and confirms the normalization $s = \operatorname{tr}\Sigma/2$.

The class $\Theta(s_0)$ consists of all parameters $\theta = (M, V, s, L, q, \mu, b, \nu)$ satisfying these conditions and $0 \le s \le s_0$. Every listed parameter is unknown. Define the physical state–return loading, transition and leverage amplitude by

$$H = AL, \qquad Z = e^{M\Delta}, \qquad e = \|L\|_F. \tag{9}$$

For a fixed $\eta > 0$, the performance subset $\Theta_\eta(s_0)$ is defined by

$$\sigma_{\min}([H, ZH]) \ge \eta \|H\|_F. \tag{10}$$

Here $\sigma_{\min}$ is the smaller row singular value, including zero. At $H = 0$ both sides vanish: zero leverage and the zero-scale boundary are included. At $H \ne 0$, the instantaneous and propagated response together span both state coordinates. The condition depends on H, so an orthogonal driver refactorization $A \mapsto AO$, $L \mapsto O^\top L$ leaves it unchanged.

The estimator's candidates range over the full class $\Theta(s_0)$, as does confidence coverage. The number η indexes precision guarantees only. Constants and the sufficiently small fixed covariance ceiling s_0 may depend on it, the grid and compact envelopes; no uniformity as η or κ tends to zero is asserted.

We use Frobenius norms for matrices and Euclidean norms for vectors. The structural loss and a fixed nonzero contrast are

$$\|\Delta(\widehat{M} - M)\|_F^2, \qquad \psi_C(M) = \operatorname{tr}(C^\top \Delta M). \tag{11}$$

For two parameter values put $s_* = \max(s, s')$ and $h_* = \max(\|\sqrt{\Delta} H\|_F, \|\sqrt{\Delta} H'\|_F)$, and define

$$\begin{aligned} d_\eta(\theta, \theta') = {} & \|\Sigma - \Sigma'\|_F + \|\sqrt{\Delta}(H - H')\|_F + (s_* + h_*)\|\Delta(M - M')\|_F \\ & + \|\Delta(b - b')\| + \|\mu - \mu'\| + \|\nu - \nu'\| + |q - q'|. \end{aligned} \tag{12}$$

No triangle property of this weighted comparison is needed. Rescaling time to $\Delta = 1$ replaces (M, H, b) by $(\Delta M, \sqrt{\Delta} H, \Delta b)$ and adds the known shift $(\log \Delta)/2$ to each state mean. We restore these units in conclusions.

3. Dynamic excitation and the information boundary

Set $\rho_n(s, e) = \min\{1, [n(s^2 + se^2)]^{-1}\}$ for $s > 0$, and set $\rho_n(0, e) = 1$.

Theorem 3.1. *For fixed $\mathcal{K}$, grid, nuisance bounds, residual margin and excitation margin, there is a sufficiently small fixed $s_0 > 0$ with the following properties.*

(A) *If $\theta \in \Theta_\eta(s_0)$ and $\theta' \in \Theta(s_0)$ have the same joint law of observed prices for $n \ge 2$, then at $s > 0$ their $(M, \Sigma, L, q, \mu, b, \nu)$ agree, and hence also s, V. At $s = 0$ only (μ, b, q, ν) remain in the law. No zero-scale law equals a positive-scale law anywhere in $\Theta(s_0)$.*

(B) *One measurable exact-moment estimator over the full $\Theta(s_0)$, not supplied s, L, η or any nuisance value, satisfies*

$$\sup_{\theta \in \Theta_\eta(s_0)} \mathbb{E}_\theta d_\eta(\widehat{\theta}_n, \theta)^2 \le C/n. \tag{13}$$

At each such truth of positive scale,

$$\mathbb{E}_\theta\|\Delta(\widehat{M}_n - M)\|_F^2 \le C\min\{1, [n(s^2+se^2)]^{-1}\}. \tag{14}$$

(C) *Suppose $\mathcal{K}$ contains $V = I$ and a full four-dimensional drift neighborhood of*

$$M_0 = \begin{pmatrix} -a_1 & g \\ g & -a_2 \end{pmatrix}, \qquad a_i > 0, \quad 0 < g < \sqrt{a_1 a_2}. \tag{15}$$

The ball and η are as in Proposition 3.2 below. For its fixed-(s,e) excited slices $\Theta_{\eta;s,e}$, for $0 < s \le s_0$ and $0 \le e \le \sqrt{2(1-\kappa)}$,

$$\inf_{\widetilde{M}} \sup_{\theta\in\Theta_{\eta;s,e}} \mathbb{E}_\theta\|\Delta(\widetilde{M} - M)\|_F^2 \asymp \rho_n(s,e). \tag{16}$$

For every fixed nonzero C, scalar contrast risk has order $\|C\|_F^2\rho_n(s,e)$. The infima range over all measurable estimators using every noisy level. For $0 \le e \le \sqrt{1-\kappa}$ the lower orders already hold on the rank-one subfamily $L = eE_{12}$. Full-rank $L = (e/\sqrt{2})I$ supplies the entire feasible Frobenius range. These subfamilies are not restrictions supplied to the estimator.

(D) *For every fixed $\alpha \in (0,1)$ there are observable confidence sets, with all nuisance parameters projected, whose coverage is at least $1-\alpha$ uniformly over full $\Theta(s_0)$, including nonexciting truths and $s = 0$. At $\theta \in \Theta_\eta(s_0)$ their projected contrast diameter satisfies*

$$\mathbb{E}_\theta \operatorname{diam}(\mathcal{I}_{n,C})^2 \le C_\alpha\|C\|_F^2\rho_n(s,e). \tag{17}$$

Under the neighborhood assumption in (C)*, for fixed $\alpha < 1/4$ this is also the worst-case lower order for every honest confidence set based on all prices.*

The theorem is an order comparison for squared loss, not an efficient information or LAN assertion. The physical H is in (13); no uniform unweighted root-n rate for L is asserted. The proofs of identification and upper inference are in Sections 4–6, and the all-price lower proof is in Section 7.

Two examples clarify why propagation enters the rate. In the coupled model below, rank-one leverage reaches both state coordinates and permits recovery of all four drift entries. In the separate diagonal model of Section 8, with known loading support, rank-one leverage accelerates inference on only one persistence parameter; the other retains its covariance-only rate. The same instantaneous rank can therefore have different consequences for structural learning.

Proposition 3.2. *The coupled example* (15)*, with $V = I$ and $L = eE_{12}$, satisfies* (10) *on a full four-dimensional drift ball with one positive normalized margin, uniformly for $0 < e \le \sqrt{1-\kappa}$. The same ball admits a positive uniform margin for $L = (e/\sqrt{2})I$, $0 < e \le \sqrt{2(1-\kappa)}$, at $Q = I$.*

Proof. By the two-dimensional Cayley–Hamilton calculus, $e^{M\Delta} = uI + vM$. For distinct eigenvalues,

$$v = \frac{e^{\lambda_1\Delta} - e^{\lambda_2\Delta}}{\lambda_1 - \lambda_2}.$$

A vanishing numerator would require their difference to be a nonzero $2\pi i/\Delta$ multiple, impossible on the strict strip (8). For a repeated eigenvalue and a nontrivial Jordan block, the quotient limit is $v = \Delta e^{\lambda\Delta} \ne 0$. A repeated, diagonalizable two-dimensional matrix is scalar, in which case propagation cannot improve loading rank. Thus for nonscalar M, $[H, ZH]$ and $[H, MH]$ have the same row rank.

At (15) the first column of $\operatorname{chol}(-2M_0)$ is

$$h_0 = \begin{pmatrix} \sqrt{2a_1} \\ -2g/\sqrt{2a_1} \end{pmatrix}, \qquad \det[h_0, M_0h_0] = \frac{2g(a_1a_2 - g^2)}{a_1} > 0.$$

Indeed M_0h_0 has entries $-a_1\sqrt{2a_1} - 2g^2/\sqrt{2a_1}$ and $g\sqrt{2a_1} + 2a_2g/\sqrt{2a_1}$. Only the second column of A_0L is nonzero, but it and its propagation span the two rows. Continuity gives a closed drift ball, full in all four entries, with negative definite symmetric part, a strict strip margin and a positive normalized singular value. The factor $\sqrt{s}e$ cancels from the normalized condition. For $L = (e/\sqrt{2})I$, A_0 is uniformly invertible on that ball, giving another positive normalized margin.

We choose η below both margins and shrink the ball if necessary. The stated amplitude ranges satisfy (6) with $Q = I$. □

This propagation algebra is classical reachability and shifted-response realization; Oymak and Ozay [5] give an LTI identification and conditioning antecedent with observed Gaussian inputs. Here the inputs and the loading are not observed. At leading order $D_k = Z^{k-1}W$, $W = F_1(M)H$. If W has rank one, choose a nonzero row $w^\top$ with $w^\top W = 0$. The perturbation $Z' = Z + uw^\top$ leaves D_1, D_2 unchanged when W is fixed. If $[W, ZW]$ has row rank two, $w^\top ZW \neq 0$, and $Z'^2W = Z^2W + uw^\top ZW$ changes the third response. A small admissible perturbation can keep W fixed by setting $M' = \Delta^{-1}\operatorname{Log} Z'$ and $H' = F_1(M')^{-1}W$. This is an ambiguity of the leading two-lag response, not an exact observed-law equivalence. Section 5 proves that the three-lag mechanism survives the nonlinear moments.

Corollary 3.3. *Under the lower-neighborhood hypothesis of Theorem 3.1, along $s_n = n^{-a}$ with $1/2 < a < 1$, zero leverage prevents uniform drift consistency. The single estimator is consistent at truths with any fixed nonzero leverage satisfying a fixed excitation margin on the same grid. For $a = 3/4$, its excited squared risk order is $n^{-1/4}$, with matching worst-case lower order on the excited slices.*

Proof. Here $ns_n^2 = n^{1-2a} \to 0$, so (16) at $e = 0$ stays bounded below. For fixed $e > 0$, $n(s_n^2 + s_n e^2) = n^{1-2a} + e^2 n^{1-a} \to \infty$, and (14) tends to zero. At $a = 3/4$ the dominating term is $e^2 n^{1/4}$. The sequences eventually lie below the fixed s_0. The first statement is minimax worst-case impossibility; it is not a pointwise lower claim at every zero-leverage drift. □

Remark 3.4. For fixed n, the joint law of all distinct noisy levels is the convolution of the corresponding clean-price law with a Gaussian having strictly positive diagonal covariance. The convolution density is everywhere positive, since the Gaussian kernel is positive at every point. Thus any two model laws are mutually absolutely continuous. If one estimator recovered two different structural values almost surely under their respective laws, the event of recovery of the first would have probability one under both laws, contradicting recovery of the second. This argument also applies to singular Brownian covariance, but is not a rate assertion at perfect leverage. The nondegenerate asymptotic theorem above uses fixed η, κ and an unevaluated small covariance ceiling. Its observable estimators and sets are existence constructions, not calibrated practical confidence widths.

4. Exact moments of observed prices

The observable moments must distinguish the drift while allowing carry, covariance, loading and level noise to vary. We derive their exact forms before comparing their sensitivity to those parameters.

For $t \in \{r, h\}$ put $R_i^t(k) = P_i(k\Delta + t) - P_i(k\Delta)$ and define

$$m_i = \mathbb{E}R_i^h = b_i h, \qquad q_i^t = \mathbb{E}(R_i^t)^2, \qquad w_i^t = q_i^t - (t/h)^2 m_i^2, \qquad \lambda_i = e^{2\mu_i + 2\Sigma_{ii}}.$$

The raw second moment q_i^t is distinct from the return correlation q. Itô isometry and independent endpoint errors give

$$w_i^t = t\lambda_i + 2\nu_i, \qquad \lambda_i = \frac{w_i^h - w_i^r}{h - r}, \qquad \nu_i = \frac{hw_i^r - rw_i^h}{2(h - r)}. \tag{18}$$

These identities do not require the return driver to be independent of the state. For $k = 1, 2, 3$ define

$$T_{kij} = \mathbb{E}[R_i^h(k)^2 R_j^h(0)] - q_i^h m_j, \qquad D_{kij} = \frac{T_{kij}}{2h^2\lambda_i\sqrt{\lambda_j}}. \tag{19}$$

For $k = 1, 2$ define

$$G_{kij} = \frac{\mathbb{E}[R_i^h(k)^2 R_j^h(0)^2] - q_i^h q_j^h - 2m_j T_{kij}}{h^2\lambda_i\lambda_j}. \tag{20}$$

An earlier linear martingale return can correlate with a later square, requiring the unknown-carry correction $2m_jT_{kij}$. A later linear return has zero mean against earlier variables.

The exact raw feature vector $F(\theta) \in \mathbb{R}^{27}$ consists of

$$\left(m_i, q_i^r, q_i^h; \mathbb{E}[R_1^h(0)R_2^h(0)]; \mathbb{E}[R_i^h(k)^2 R_j^h(0)]_{k=1,2,3}; \mathbb{E}[R_i^h(k)^2 R_j^h(0)^2]_{k=1,2}\right), \tag{21}$$

with $i, j = 1, 2$ in the mixed arrays. The counts are $6+1+12+8 = 27$. Estimation and inference use this exact feature map. The transformations (18)–(20) are Lipschitz on its compact image: λ_i is uniformly bounded above and away from zero. Shared endpoint errors between R_i^r and R_i^h are retained in empirical dependence; no independence between those returns is assumed.

Write, for $t \geq 0$,

$$C(t) = e^{Mt}\Sigma, \qquad K(t) = e^{Mt}H. \tag{22}$$

In particular $K_{ij}(t-u)du = \mathrm{Cov}(Y_i(t), dU_j(u))$ for $t > u$. The symbol $C(t)$ here is a covariance kernel, not the fixed contrast matrix of (11).

Lemma 4.1. *For $z = k\Delta + t$, $\tau = z - u$,*

$$D_{kij} = \frac{1}{h^2}\int_0^h \int_0^h K_{ij}(\tau) e^{2C_{ij}(\tau) - \Sigma_{jj}/2}\, du\, dt. \tag{23}$$

Moreover $G_k = G_k^0 + J_k$, where

$$G_{kij}^0 = \frac{1}{h^2}\int_0^h \int_0^h [e^{4C_{ij}(z-u)} - 1]\, du\, dt \tag{24}$$

and

$$\begin{aligned} J_{kij} = \frac{4}{h^2}\int_0^h \int_0^h \int_0^u & K_{ij}(z-u)[K_{jj}(u-v) + 2K_{ij}(z-v)] \\ & \cdot e^{-\Sigma_{jj} + C_{jj}(u-v) + 2C_{ij}(z-u) + 2C_{ij}(z-v)}\, dv\, du\, dt. \end{aligned} \tag{25}$$

The same-block cross moment satisfies

$$\mathbb{E}[R_1^h(0)R_2^h(0)] - m_1 m_2 = hq\sqrt{\lambda_1\lambda_2}\, e^{\Sigma_{12} - (\Sigma_{11}+\Sigma_{22})/2}. \tag{26}$$

Proof. We first establish the Gaussian exponential-tilting identity needed here. Let Z_0 be a Gaussian linear functional of the two-sided drivers. For an adapted integrand f on a bounded interval, whenever the expressions are integrable,

$$\mathbb{E}\left[e^{Z_0}\int f(u) dU_j(u)\right] = \int \frac{\mathrm{Cov}(dU_j(u), Z_0)}{du}\, \mathbb{E}[e^{Z_0} f(u)]\, du. \tag{27}$$

After division by $\mathbb{E}e^{Z_0}$, the left-hand side is an expectation under the Gaussian tilt $\exp(Z_0 - \mathbb{E}Z_0 - \mathrm{Var}(Z_0)/2)$. The driver covariances are unchanged, and their means shift by their covariance with Z_0. Centered increments remain independent of the preceding sigma-field. For an adapted step process, the centered stochastic sum therefore has zero expectation, leaving exactly the deterministic drift shift in (27). Localization, Hölder and Itô convergence extend it to the integrands used below. The Gaussian state has uniformly bounded fixed exponential moments, and the Burkholder–Davis–Gundy (BDG) inequality supplies fixed return moments, so both sides converge in the required L^1 spaces.

Let $N_j(u) = \int_0^u e^{Y_j(v)} dU_j(v)$ be the clean centered earlier return. A squared later martingale increment, when multiplied by an earlier measurable variable, can be replaced in expectation by its quadratic-variation integral. This follows from Itô's formula for its square, since the remaining later stochastic integral has zero expectation. For $z \geq \Delta > h$, (27) with $Z_0 = 2Y_i(z)$ gives

$$\mathbb{E}[e^{2Y_i(z)} N_j(h)] = 2\int_0^h K_{ij}(z-u)\, \mathbb{E}[e^{Y_j(u) + 2Y_i(z)}]\, du.$$

The Gaussian expectation is

$$\mathbb{E}[e^{Y_j(u) + 2Y_i(z)}] = \lambda_i\sqrt{\lambda_j}\, e^{2C_{ij}(z-u) - \Sigma_{jj}/2}.$$

Integrating in $z = k\Delta + t$ proves (23) for the clean centered returns.

For the fourth moment, Itô's formula gives

$$N_j(h)^2 = \int_0^h e^{2Y_j(u)}du + 2\int_0^h N_j(u)e^{Y_j(u)}dU_j(u).$$

Against $e^{2Y_i(z)}$ the first term has expectation $\lambda_i\lambda_j e^{4C_{ij}(z-u)}$, producing (24) after subtracting the product of second moments. Applying (27) to the second term gives

$$4\int_0^h K_{ij}(z-u)\mathbb{E}[N_j(u)e^{Y_j(u)+2Y_i(z)}]du.$$

A second application, now with $Z_0 = Y_j(u) + 2Y_i(z)$, gives

$$\mathbb{E}[N_j(u)e^{Y_j(u)+2Y_i(z)}] = \int_0^u [K_{jj}(u-v) + 2K_{ij}(z-v)]\mathbb{E}[e^{Y_j(v)+Y_j(u)+2Y_i(z)}]dv.$$

Dividing the last Gaussian exponential expectation by $\lambda_i\lambda_j$ yields exactly

$$e^{-\Sigma_{jj}+C_{jj}(u-v)+2C_{ij}(z-u)+2C_{ij}(z-v)}.$$

This proves the full triple integral (25).

The later and earlier observed block returns are, respectively, $m_i + N_i + \xi_i$ and $m_j + N_j + \xi_j$. The disjoint block errors ξ_i, ξ_j are independent centered Gaussians and independent of the clean paths. The later centered return has zero mean against any earlier variable, and disjoint clean martingale returns have zero cross second moment. Expanding the cubic thus leaves $\mathbb{E}[N_i^2N_j]$ after subtracting $q_i^h m_j$. Expanding the raw fourth moment leaves $\mathbb{E}[N_i^2N_j^2] - \mathbb{E}N_i^2\mathbb{E}N_j^2$ and $2m_j\mathbb{E}[N_i^2N_j]$ after subtracting $q_i^h q_j^h$. The latter is precisely the carry correction in (20). Error variances in the two disjoint blocks cancel in these corrected moments. This argument does not cancel shared errors within a block.

Finally the cross quadratic variation is $q\int_0^h e^{Y_1(t)+Y_2(t)}dt$. Independent errors across the two coordinates add no cross covariance. The Gaussian expectation $e^{\mu_1+\mu_2+(\Sigma_{11}+\Sigma_{22})/2+\Sigma_{12}}$ proves (26). Every integration is over a bounded interval; Hölder, Gaussian exponential moments and BDG justify Fubini and the localization limits above. In particular, we have not assumed a fixed-mean conditional-normal return likelihood under leverage. □

These formulas exhibit $D_k = O(\sqrt{s}e)$, $G_k^0 = O(s)$ and $J_k = O(se^2)$. The leverage response order is classical [6, 3]; neither it nor a moment rank calculation proves the all-price information order.

5. A one-endpoint inverse

We work in $\Delta = 1$ units in this section. Write $\delta_M = \|M - M'\|_F$, $\delta_\Sigma = \|\Sigma - \Sigma'\|_F$, $\delta_H = \|H - H'\|_F$, and $\epsilon = \|F(\theta) - F(\theta')\|$. Until excitation is explicitly invoked, both points can be anywhere in the admitted compact class. By admissibility and bounded A_0,

$$h_*^2 \le Cs_*, \qquad h_* = \max(\|H\|_F, \|H'\|_F), \quad s_* = \max(s, s'). \tag{28}$$

All constants below are independent of the true s, e, n.

5.1. The covariance-pair inverse.

Define the matrix averaging kernels

$$F_k(M) = \frac{1}{h^2}\int_0^h\int_0^h e^{M(k\Delta+t-u)}du\,dt, \quad A_h(M) = \frac{1}{h}\int_0^h e^{Mt}dt. \tag{29}$$

They obey $F_k = Z^{k-1}F_1$ and $F_1 = A_h(M)ZA_h(-M)$. Each factor is invertible on the drift class. The scalar entire function $(e^{hz}-1)/(hz)$ has value one at zero and only nonzero purely imaginary zeros; no Hurwitz eigenvalue, nor its negative, can be such a zero. Functional calculus proves invertibility even at a Jordan block, and compactness bounds F_1, F_1^{-1} uniformly.

On the compact nonaliasing image,

$$M = \Delta^{-1}\operatorname{Log} Z, \qquad \|M - M'\|_F \le C\|Z - Z'\|_F. \tag{30}$$

Here Z has spectrum off the nonpositive real axis by (8), and the principal logarithm of $e^{M\Delta}$ equals $M\Delta$, also for defective M. The logarithm is analytic in a neighborhood of each image

point. A finite covering bounds its local Fréchet derivatives and gives the Lipschitz inequality for close pairs. For pairs outside those neighborhoods, continuity, injectivity and compact separation give a positive minimum image distance, while the drift diameter is bounded. This proves the global inequality; no spectral diagonalization assumption is used.

Lemma 5.1. *For a sufficiently small fixed covariance ceiling,*

$$\delta_\Sigma + s_*\delta_M \le C\|(G_1^0, G_2^0) - (G_1^{0\prime}, G_2^{0\prime})\|. \tag{31}$$

Proof. The linear terms in (24) are $B_1 = 4F_1\Sigma$ and $B_2 = ZB_1$. We may label the endpoints so that $s' = s_* > 0$. Then $\sigma_{\min}(B_1') \ge cs_*$, and

$$(Z - Z')B_1' = (B_2 - B_2') - Z(B_1 - B_1')$$

controls $s_*\|Z - Z'\|_F$, hence $s_*\delta_M$ by (30), by the linear-pair difference. Also

$$B_1 - B_1' = 4F_1(\Sigma - \Sigma') + 4(F_1 - F_1')\Sigma'$$

and bounded F_1^{-1} control δ_Σ by that difference and $s_*\delta_M$. If $s_* = 0$ both terms to be bounded vanish.

For the exponential remainder $e^{4C} - 1 - 4C$, finite-time differentiation has covariance derivative $O(s_*)$ and drift derivative $O(s_*^2)$. Thus its integrated pair difference is at most $Cs_*[\delta_\Sigma + s_*\delta_M]$. These estimates can be checked along bounded matrix interpolations: finite-time exponentials have bounded Fréchet derivatives regardless of stability of the interpolation, and $\|\Sigma\| \le Cs_*$ throughout. This does not require convexity of $\mathcal{K}$. Applying the linear inverse to the nonlinear pair gives a remainder that can be absorbed when s_0 satisfies $Cs_0 \le 1/2$. □

5.2. **Profiling the unknown physical loading.**

Lemma 5.2. *Write $D_k = F_kH + E_k$, $k = 1, 2, 3$. Uniformly,*

$$\|E_k\|_F \le Cs\|H\|_F,$$
$$\|E_k - E_k'\|_F \le C[s_*\delta_H + h_*\delta_\Sigma + s_*h_*\delta_M]. \tag{32}$$

For a sufficiently small fixed s_0, profiling H through D_1 gives

$$\delta_H \le C[\epsilon + h_*\delta_M + h_*\delta_\Sigma],$$
$$\|J_k - J_k'\|_F \le C[h_*\epsilon + h_*^2(\delta_M + \delta_\Sigma)]. \tag{33}$$

In addition, with $a = \|H\|_F$ at the unprimed endpoint,

$$h_* \le C(a + \epsilon). \tag{34}$$

No excitation of the primed endpoint is needed.

Proof. Subtracting one from the exponential in (23) leaves a remainder of order $O(s)$. Its covariance derivative is bounded, and its drift derivative is $O(s)$, on all the finite integration times involved. The expression is linear in H, while the derivative of e^{Mt} is bounded at finite times. Taking differences gives (32). Bounded interpolations as above justify these pairwise bounds on a possibly nonconvex compact parameter class.

For each return column j the exact first cubic is $D_{1,:j} = \mathcal{B}_j(M, \Sigma)H_{:j}$, where

$$(\mathcal{B}_j)_{ip} = \frac{1}{h^2}\int_0^h\int_0^h [e^{M(\Delta+t-u)}]_{ip}e^{2C_{ij}(\Delta+t-u)-\Sigma_{jj}/2}\,du\,dt. \tag{35}$$

It is $F_1 + O(s)$. The preceding uniform inverse bound and a Neumann argument give uniformly bounded inverses for small s_0. Their coefficient functions are Lipschitz: locally this follows by differentiation of an inverse, and globally by compact separation for distant pairs. Therefore subtracting $H_{:j} = \mathcal{B}_j^{-1}D_{1,:j}$ gives the first bound in (33). Here differences of D_1 are bounded by $C\epsilon$ through the raw normalization, and $\|D_1\|_F \le C\|H\|_F$.

Substitution of the column profiles into (25) gives $\overline{J}_k(M, \Sigma, D_1)$ as a quadratic polynomial in the observable D_1, with bounded Lipschitz coefficient functions. Its difference is bounded by a linear term $Ch_*\epsilon$ and a coefficient difference $Ch_*^2(\delta_M + \delta_\Sigma)$, proving the second bound. Finally

$$\|H'\|_F \le C\|D_1'\|_F \le C(\|D_1\|_F + \epsilon) \le C(a + \epsilon),$$

which proves (34). In particular, an arbitrary alternative loading is controlled by the observed first cubic, not assumed known or excited. □

Lemma 5.3. *For arbitrary admitted endpoints,*

$$\delta_\Sigma \le C[\epsilon + s_*\delta_M]. \tag{36}$$

Proof. From (24), $G_1^0 = 4F_1\Sigma + R_1^0$, with

$$\|R_1^0 - R_1^{0\prime}\|_F \le C[s_*\delta_\Sigma + s_*^2\delta_M]. \tag{37}$$

We take the difference of $G_1 = G_1^0 + J_1$ between the endpoints and apply the inverse of F_1. The difference of F_1 contributes at most $Cs_*\delta_M$. By (33), (28) and bounded h_*, the correction contributes at most $C\epsilon + Cs_*\delta_M + Cs_*\delta_\Sigma$. Together with (37) this yields $\delta_\Sigma \le C[\epsilon + s_*\delta_M] + Cs_0\delta_\Sigma$. For fixed $Cs_0 \le 1/2$, the last term can be absorbed. The covariance derivative of the profiled correction is small in s, so this step requires no smallness assumption on L. □

5.3. **Lag three and excitation at only one endpoint.**

Lemma 5.4. *For every $\theta \in \Theta_\eta(s_0)$ and every $\theta' \in \Theta(s_0)$, after another fixed smallness choice for s_0,*

$$h_*\delta_M \le C_\eta\epsilon. \tag{38}$$

Proof. The dynamic lag residuals $\mathcal{R}_k = E_{k+1} - ZE_k$, $k = 1, 2$, satisfy the exact identities

$$D_2 = ZD_1 + \mathcal{R}_1, \qquad D_3 = ZD_2 + \mathcal{R}_2. \tag{39}$$

By (32),

$$\|\mathcal{R}_k - \mathcal{R}_k'\|_F \le C[s_*\delta_H + h_*\delta_\Sigma + s_*h_*\delta_M].$$

The additional term $(Z - Z')E_k'$ has precisely the last order here. The first bound of (33), together with bounded s_*, gives

$$\|\mathcal{R}_k - \mathcal{R}_k'\|_F \le C[s_*\epsilon + h_*\delta_\Sigma + s_*h_*\delta_M]. \tag{40}$$

If $\epsilon \ge a = \|H\|_F$, then (34) gives $h_* \le C\epsilon$. The compact drift diameter proves (38) immediately. This case includes $a = 0$, so it includes zero leverage and zero scale without any assumption on the alternative.

Suppose instead $\epsilon < a$. Then $a > 0$ and $h_* \le Ca$. At the unprimed endpoint only, excitation gives

$$\begin{aligned} \sigma_{\min}([D_1, D_2]) &\ge \sigma_{\min}(F_1[H, ZH]) - Csa \\ &\ge c\eta a - Csa \ge c_\eta a, \end{aligned} \tag{41}$$

after choosing $s_0 \le c\eta/(2C)$. We used that F_1 commutes with Z, hence $[F_1H, F_2H] = F_1[H, ZH]$. The difference of (39) takes the form

$$(Z - Z')D_1 = (D_2 - D_2') - Z'(D_1 - D_1') - (\mathcal{R}_1 - \mathcal{R}_1'),$$

and similarly with D_2, D_3. Equations (40) and $h_* \le Ca$ give

$$\|(Z - Z')[D_1, D_2]\|_F \le C[\epsilon + a\delta_\Sigma + s_*a\delta_M]. \tag{42}$$

For any matrix T, (41) implies $c_\eta a\|T\|_F \le \|T[D_1, D_2]\|_F$. Together with (30), (42) and (36) and bounded a, this gives

$$a\delta_M \le C_\eta\epsilon + C_\eta s_*a\delta_M.$$

Choosing $C_\eta s_0 \le 1/2$ allows the last term to be absorbed. Since $h_* \le Ca$ in this case, (38) follows. The argument places no lower bound on the singular values of $[H', Z'H']$. □

Proposition 5.5. *For every excited truth and every admitted alternative,*

$$d_\eta(\theta, \theta') \le C_\eta\|F(\theta) - F(\theta')\|. \tag{43}$$

Proof. Lemma 5.1, applied to $G^0 = G - J$, together with the profiled correction and Lemma 5.4, gives

$$\delta_\Sigma + s_*\delta_M \le C_\eta\epsilon + Ch_*^2\delta_\Sigma.$$

Indeed the term $h_*^2\delta_M$ is bounded by $Ch_*\epsilon$, and h_* is uniformly bounded. By (28), a further fixed smallness choice allows $Ch_*^2\delta_\Sigma$ to be absorbed. Then (33) and (38) yield

$$\delta_\Sigma + \delta_H + (s_* + h_*)\delta_M \le C_\eta\epsilon. \tag{44}$$

The regular nuisance parameters follow explicitly: $b_i = m_i/h$, ν_i, λ_i come from (18), and $\mu_i = (\log\lambda_i)/2 - \Sigma_{ii}$. Finally, by (26),

$$q = \frac{\mathbb{E}[R_1^h R_2^h] - m_1 m_2}{h\sqrt{\lambda_1\lambda_2}} e^{-\Sigma_{12} + (\Sigma_{11}+\Sigma_{22})/2}.$$

These maps are uniformly Lipschitz on the compact moment image. Restoring the Δ units specified after (12) gives (43).

If $s > 0$, equality of features gives M, Σ, H and all regular nuisance parameters. Then $s = \operatorname{tr}\Sigma/2$, $V = \Sigma/s$, A is determined by (4), and $L = A^{-1}H$. At zero scale the price law is that of a constant-carry Brownian log price with covariance $\operatorname{diag}(e^\mu)Q\operatorname{diag}(e^\mu)$ and the given level noise, independent of M, V, L. The regular moments identify its retained parameters. A zero-scale endpoint is always in $\Theta_\eta(s_0)$. Taking it as the unprimed endpoint in (43) rules out equality with any positive-scale law. Equality of the joint laws of observed prices implies equality of the available features for $n \ge 2$. Conversely equal retained parameters construct equal laws. This proves Theorem 3.1(A). It does not classify pairs of large nonexciting positive-scale models. □

6. Adaptive estimation and honest sets

The inverse bound converts moment error into drift error according to the signal strength at the truth. Estimation over the full parameter class then allows that strength to be unknown, while confidence coverage remains valid even where the drift cannot be precisely estimated.

Lemma 6.1. *The average of the 27 raw features over origins $k = 0, \dots, n-2$, with all block indices in* (21) *shifted accordingly, is denoted by $\widehat{F}_n$. A deterministic compact-class envelope C_F satisfies*

$$\sup_{\theta\in\Theta(s_0)} \mathbb{E}_\theta\|\widehat{F}_n - F(\theta)\|^2 \le C_F/n. \tag{45}$$

This includes nonexciting leverage and zero scale.

Proof. An origin uses at most four neighboring blocks through lag three, all already in $\mathcal{T}_n$. There are $n - 1 \ge n/2$ origins. Every feature is a polynomial of degree at most four in returns. For every fixed p, BDG bounds a clean return by the $p/2$ moment of $\int e^{2Y_i(t)}dt$ on a bounded interval. Hölder and Gaussian exponential moments bound this uniformly; the bounded carry and Gaussian errors preserve the bound. Thus all raw features have uniformly bounded fixed L^p norms.

For $s > 0$, the normalized state $s^{-1/2}V^{-1/2}(Y - \mu)$ has transition correlation matrix $R_t = V^{-1/2}e^{Mt}V^{1/2}$, with $\|R_t\|_{\mathrm{op}} \le Ce^{-ct}$ independent of s. For sufficiently large t, its pair of standardized states has joint covariance $\left(\begin{smallmatrix} I & R_t^\top \\ R_t & I \end{smallmatrix}\right)$. Against the product Gaussian law the trace term in the Gaussian KL formula vanishes and the determinant term gives

$$\mathrm{KL} = -\tfrac{1}{2}\log\det(I - R_tR_t^\top) \le Ce^{-2ct}.$$

Pinsker therefore bounds the corresponding total variation by Ce^{-ct}. For a stationary Markov state, conditioning past and future on their boundary states and then applying data processing passes this bound to absolute regularity of separated state segments. It also passes to emission blocks: after the boundary, joint Brownian innovations are independent of the past, even though state and return innovations within the future are correlated by L. Separated observation-error

sets are independent. Hence the feature blocks have an exponential absolute- regularity bound once their spans $3\Delta + h$ are separated.

One separated block can be coupled to an independent copy with mismatch probability at most its absolute-regularity coefficient β. For $p > 2$, Hölder on the mismatch event gives

$$|\operatorname{Cov}(f, g)| \le C\beta^{1-2/p}\|f\|_p\|g\|_p.$$

Indeed independence makes the covariance with the copy zero, and the difference $g - g^*$ is supported on the mismatch event; the three Hölder exponents are $p, p, (1 - 2/p)^{-1}$. Summing the exponential covariance tail and controlling the finitely many overlapping offsets by Cauchy–Schwarz yields a uniform absolute covariance sum. At $s = 0$ separated features depend on independent return innovations and errors and have finite-range dependence. For an average of $n - 1$ origins, the double covariance sum is at most $C(n - 1)$, proving (45) after summing over the 27 coordinates. The fixed moment envelopes, overlap count and geometric tail give a known deterministic upper envelope C_F; no long-run variance estimate or volatility proxy is required. □

Let $(\theta_j)_{j\ge1}$ be a fixed enumeration of a countable dense subset of full $\Theta(s_0)$. We define $\widehat{\theta}_n$ as the first enumerated point satisfying

$$\|F(\theta_j) - \widehat{F}_n\| \le \inf_{l\ge1} \|F(\theta_l) - \widehat{F}_n\| + 1/n. \tag{46}$$

Continuity of the exact moment map makes the countable infimum equal to the compact infimum. The first-index selection is measurable, since all its inequalities involve measurable real functions and a countable infimum. It exists by the definition of the infimum, and

$$\|F(\widehat{\theta}_n) - F(\theta)\| \le 2\|\widehat{F}_n - F(\theta)\| + 1/n. \tag{47}$$

At an excited truth, Proposition 5.5 applies to the selected candidate regardless of its excitation. Lemma 6.1 then proves (13). Uniform bounds for A_0, A_0^{-1} give

$$c\sqrt{s}e \le \|\sqrt{\Delta}H\|_F \le C\sqrt{s}e. \tag{48}$$

The drift weight in (12) is at least $s + \|\sqrt{\Delta}H\|_F$. Its square is comparable, within fixed constants, to $s^2 + se^2$. Dividing the joint bound by the squared truth weight and using the compact drift diameter gives (14). This proves Theorem 3.1(B). The construction gives an observable, measurable estimator based on the exact moment map over the full parameter class. Its computational efficiency is not established.

For fixed $\alpha \in (0, 1)$ define the full-class moment confidence set

$$\mathcal{C}_{n,\alpha} = \left\{\vartheta \in \Theta(s_0) : \|\widehat{F}_n - F(\vartheta)\| \le \sqrt{C_F/(\alpha n)}\right\}. \tag{49}$$

Projection gives a drift set or the contrast set $\mathcal{I}_{n,C} = \{\psi_C(M_\vartheta) : \vartheta \in \mathcal{C}_{n,\alpha}\}$, with zero diameter for an empty set. The continuous compact construction makes these observable sets and their diameters measurable. Markov's inequality applied to (45) gives

$$\inf_{\theta\in\Theta(s_0)} \mathbb{P}_\theta\{\theta \in \mathcal{C}_{n,\alpha}\} \ge 1 - \alpha. \tag{50}$$

This coverage includes every nonexciting truth and $s = 0$. Projection removes nuisance coordinates rather than treating them as known, as in identification-robust inference [2, 1].

At an excited truth, every candidate in (49) obeys, even off the coverage event,

$$\|F(\vartheta) - F(\theta)\| \le Z_n := \sqrt{C_F/(\alpha n)} + \|\widehat{F}_n - F(\theta)\|.$$

The one-endpoint inverse bounds its dimensionless drift distance from the truth by $C_\eta Z_n/(s + \|\sqrt{\Delta}H\|_F)$ at positive scale. Two candidates thus have contrast distance at most twice this bound times $\|C\|_F$. Since $\mathbb{E}Z_n^2 \le C_\alpha/n$, and the projection also has the fixed compact diameter, (17) follows. At zero scale the bounded diameter still gives the stated bound; structural uncertainty generally cannot shrink there. This proves the upper part of Theorem 3.1(D) without a leverage or rank pretest.

7. The information in all observed prices

The moment estimator attains the proposed rate. To show that using other features of the prices cannot improve its worst-case order, we bound the separation between nearby drift models in the full observed-price law. For dominated laws we use $\mathsf{H}^2(P,P') = \int(\sqrt{p}-\sqrt{p'})^2$, with $\mathrm{TV}(P,P') = \frac{1}{2}\int|p-p'| \le \mathsf{H}(P,P')$. The path parameter in this section is ξ, distinct from confidence error probability α.

Lemma 7.1. *Fix $Q = I$, b, μ, ν, s, e and $\overline{L}$ with $\|\overline{L}\|_F = 1$, and let $L = e\overline{L}$ satisfy $e^2\|\overline{L}\|_{\mathrm{op}}^2 \le 1-\kappa$. Suppose (M_ξ, V_ξ) is a C^2 path on a compact interval, with bounded first and second derivatives and uniform class and stability bounds on a neighborhood. The complete noisy-level laws satisfy*

$$\mathsf{H}^2(P_\xi^{(n)}, P_{\xi'}^{(n)}) \le Cn(s^2+se^2)|\xi-\xi'|^2. \tag{51}$$

The constant is uniform in n, s, e under the fixed envelopes. No excitation is needed. This is a structural-submodel bound, not a full-nuisance equivalence of experiments.

The proof is given in the next four subsections. At $s = 0$ the laws on this submodel do not depend on ξ. We therefore carry out the differentiation for $s > 0$.

7.1. **A parameter-free common driver and hidden states.** Let $\tau^2 = 1-\kappa/2$ and $r_0 = \kappa/2$. Independent standard two-dimensional two-sided Brownian motions W^0, B^0, W^1 give the representation

$$U = \tau W^0 + \sqrt{r_0}W^1, \qquad B = \tau^{-1}LW^0 + \Gamma_e^{1/2}B^0, \qquad \Gamma_e = I - LL^\top/\tau^2. \tag{52}$$

Since $\|L\|_{\mathrm{op}}^2 \le 1-\kappa$, $\Gamma_e \succeq \kappa/(2-\kappa)I$. Direct multiplication gives $\mathrm{Cov}(dU) = I\,dt$, $\mathrm{Cov}(dB) = I\,dt$ and $\mathrm{Cov}(dB, dU) = L\,dt$. The factorization preserves the law and the physical Cholesky convention, with positive independent conditional components for both state and returns.

Let I_ℓ denote the intervals between successive distinct grid times. These include every gap. Their lengths lie in

$$d_- := \min(\delta, \Delta - h) > 0, \qquad d_+ := \max(\delta, \Delta - h) < \infty, \tag{53}$$

and their number is $(n+2)N + n + 1 \le Cn$. We consider an enlarged experiment in which W^0 is observed on the full stationary prehistory up to the final observation time, together with every clean asset-by-interval increment. Both state coordinates remain hidden. The law of this additional driver is independent of ξ. Starting from the fixed clean initial levels, cumulative summation of these increments reconstructs every clean level. Adding one original independent Gaussian error for each coordinate and distinct time reconstructs all observations, including the initial noisy level and gap observations. This parameter-independent kernel reproduces shared endpoint-error dependence in any subsequently formed returns.

Given W^0, the conditional state mean is

$$m_\xi(t) = \mu + \frac{\sqrt{s}e}{\tau}\int_{-\infty}^t e^{M_\xi(t-u)}A_{0,\xi}\overline{L}\,dW^0(u), \tag{54}$$

and its covariance is sR_ξ, where, for $t \ge u$,

$$\begin{aligned} R_\xi(t,u) &= e^{M_\xi(t-u)}S_\xi, \\ S_\xi &= \int_0^\infty e^{M_\xi v}A_{0,\xi}\Gamma_e A_{0,\xi}^\top e^{M_\xi^\top v}dv. \end{aligned} \tag{55}$$

For $t < u$, the kernel is obtained by transposition. Smoothness of Cholesky on the uniformly positive diffusion cone and Duhamel's formula give

$$\|\dot{R}_\xi(t,u)\|_F \le C(1+|t-u|)e^{-c|t-u|}. \tag{56}$$

Indeed $\partial_\xi e^{M_\xi v} = \int_0^v e^{M_\xi(v-w)}\dot{M}_\xi e^{M_\xi w}dw$ is bounded by Cve^{-cv} after reducing c if necessary; differentiating the integral for S_ξ is justified by this integrable bound. In general $\dot{R}_\xi(t,t) \ne 0$, even when the unconditional covariance is fixed: conditioning on the common driver changes the covariance.

Differentiating the adapted convolution (54) gives, for every fixed p,

$$\sup_t \|\dot m_\xi(t)\|_{L^p} \le C_p\sqrt{s}e. \tag{57}$$

For example, with $x = m_\xi - \mu$,

$$dx = M_\xi x\,dt + \frac{\sqrt{s}e}{\tau}A_{0,\xi}\overline{L}\,dW^0, \quad d\dot m_\xi = (M_\xi \dot m_\xi + \dot M_\xi x)dt + \frac{\sqrt{s}e}{\tau}\dot A_{0,\xi}\overline{L}\,dW^0.$$

This stable triangular linear system has stationary second moments of order se^2, by the exponential convolution bounds. Gaussian moments give all fixed p. The second parameter derivative is likewise bounded in L^2, using the C^2 coefficients and differentiated kernels. For each finite collection of time nodes, Fubini bounds the H^2 norm in ξ almost surely; the one-dimensional embedding $H^2 \subset C^1$ gives a continuously differentiable version at these nodes.

Let j now index an asset $i(j)$ and a complete-grid interval I_j; there are $m \le Cn$ such emissions. Conditional on Y, W^0 they are independent normals with

$$a_j = b_{i(j)}|I_j| + \tau\int_{I_j} e^{Y_{i(j)}(t)}dW^0_{i(j)}(t), \qquad d_j = r_0\int_{I_j} e^{2Y_{i(j)}(t)}dt. \tag{58}$$

This follows from independence of W^1 from (Y, W^0) and independent W^1 coordinates on disjoint intervals. The random mean is a measurable Itô functional of the state and the additional observed driver, defined by its left sums.

7.2. **Finite Gaussian differentiation and the full score.** For both assets, we use a common deterministic refinement of the full grid and left sums with positive cell lengths. At a node u, let w_u be its cell length, z_u the appropriate W^0 increment, and y_u the corresponding left state value. Both coordinates at the union of physical nodes form one Gaussian vector. Distinct asset coordinates at one time are distinct Gaussian nodes. For emission j the finite quadrature is

$$a_j = b_i|I_j| + \tau\sum_{u\in j} e^{y_u}z_u, \qquad d_j = r_0 D_j, \qquad D_j = \sum_{u\in j} w_u e^{2y_u}. \tag{59}$$

Its node derivatives are

$$\mathfrak{m}_u = \partial_{y_u}a_j = \tau e^{y_u}z_u, \qquad \mathfrak{d}_u = \partial_{y_u}d_j = 2r_0 w_u e^{2y_u}.$$

Their own second derivatives are $\mathfrak{m}_u$ and $2\mathfrak{d}_u$. The state-mean derivative $\dot m_u$ is different from $\mathfrak{m}_u$.

For the one-dimensional normal density $k_j(x; a_j, d_j)$, set $Z_j = (x - a_j)/\sqrt{d_j}$, $M_j = k_{j,a}/k_j$ and $S_j = k_{j,d}/k_j$. Direct differentiation of its exponential gives

$$M_j = \frac{H_1(Z_j)}{\sqrt{d_j}}, \quad S_j = \frac{H_2(Z_j)}{2d_j}, \quad \frac{k_{j,aa}}{k_j} = \frac{H_2(Z_j)}{d_j}, \quad \frac{k_{j,ad}}{k_j} = \frac{H_3(Z_j)}{2d_j^{3/2}}, \quad \frac{k_{j,dd}}{k_j} = \frac{H_4(Z_j)}{4d_j^2}. \tag{60}$$

Here $H_1(z) = z$, $H_2(z) = z^2 - 1$, $H_3(z) = z^3 - 3z$ and $H_4(z) = z^4 - 6z^2 + 3$ are probabilists' Hermite polynomials. For a finite Gaussian vector of mean m_ξ and covariance sR_ξ,

$$\partial_\xi \mathbb{E}f(Y) = \sum_u \dot m_u \mathbb{E}f_u(Y) + \frac{s}{2}\sum_{u,v}\dot R_{uv}\mathbb{E}f_{uv}(Y). \tag{61}$$

The identity follows by differentiating the Gaussian characteristic function. In the density, the mean derivative corresponds to $-\sum_u \dot m_u \partial_u\phi$, and the covariance derivative to $(s/2)\sum_{u,v}\dot R_{uv}\partial_{uv}\phi$. Integrating against a compactly supported smooth f and integrating by parts once or twice gives (61). Equivalently these identities follow directly by differentiating the finite Gaussian density. The conditional covariance at the distinct nodes is positive definite: Γ_e and the state diffusion are positive, as are nonzero-time OU innovation covariances. For our positive emission functions, we first localize in the Gaussian nodes. The uniform moment and inverse-moment bounds proved below justify removal of that localization. Thus the identity applies to the finite Gaussian vector at each quadrature.

Let $k = \prod_j k_j$. The score coefficients are

$$\begin{aligned} \mathsf{A}_j &= \sum_{u\in j} \dot{m}_u \mathfrak{m}_u, \quad \mathsf{B}_j = \sum_{u\in j} \dot{m}_u \mathfrak{d}_u, \\ \mathsf{C}_j &= \sum_{u\in j} \dot{R}_{uu} \mathfrak{m}_u, \quad \mathsf{E}_j = \sum_{u\in j} \dot{R}_{uu}(2\mathfrak{d}_u), \end{aligned} \tag{62}$$

and

$$\begin{aligned} \mathsf{C}^{mm}_{jk} &= \sum_{u\in j, v\in k} \dot{R}_{uv}\mathfrak{m}_u\mathfrak{m}_v, \quad \mathsf{C}^{md}_{jk} = \sum_{u\in j, v\in k} \dot{R}_{uv}\mathfrak{m}_u\mathfrak{d}_v, \\ \mathsf{C}^{dm}_{jk} &= \sum_{u\in j, v\in k} \dot{R}_{uv}\mathfrak{d}_u\mathfrak{m}_v, \quad \mathsf{C}^{dd}_{jk} = \sum_{u\in j, v\in k} \dot{R}_{uv}\mathfrak{d}_u\mathfrak{d}_v. \end{aligned} \tag{63}$$

The conditional marginal density $p_\xi(x \mid W^0)$ has derivative

$$\partial_\xi p_\xi = \mathbb{E}[k\mathcal{S} \mid W^0], \qquad \mathcal{S} = \sum_j (\mathsf{A}_j M_j + \mathsf{B}_j S_j) + \frac{s}{2}\left\{ \sum_j (\mathsf{C}_j M_j + \mathsf{E}_j S_j) + \mathcal{Q} \right\}, \tag{64}$$

where the complete quadratic part is

$$\begin{aligned} \mathcal{Q} = &\sum_j \left(\mathsf{C}^{mm}_{jj} \frac{k_{j,aa}}{k_j} + 2\mathsf{C}^{md}_{jj} \frac{k_{j,ad}}{k_j} + \mathsf{C}^{dd}_{jj} \frac{k_{j,dd}}{k_j} \right) \\ &+ 2\sum_{j<k} \left(\mathsf{C}^{mm}_{jk} M_j M_k + \mathsf{C}^{md}_{jk} M_j S_k + \mathsf{C}^{dm}_{jk} S_j M_k + \mathsf{C}^{dd}_{jk} S_j S_k \right). \end{aligned} \tag{65}$$

For a single node, $\partial_u k = k(\mathfrak{m}_u M_j + \mathfrak{d}_u S_j)$. For $u = v$, its second derivative has the additional terms $\mathfrak{m}_u k_{j,a} + 2\mathfrak{d}_u k_{j,d}$, giving $\mathsf{C}_j, \mathsf{E}_j$. The product of first emission derivatives gives the four pair types in (63); within one emission the two mixed types coincide after summing, and between two emissions symmetry of $\dot{R}$ doubles the $j < k$ sum. The mean part of (61) gives $\mathsf{A}_j, \mathsf{B}_j$. The formula therefore includes derivatives of both the random state means and the conditional covariance diagonals.

Since the law of the additional observed driver is parameter-free, conditional Cauchy–Schwarz (or Jensen for the posterior score) bounds joint marginal Fisher information by

$$I_\xi \le \mathbb{E} \int k\mathcal{S}^2 \, dx. \tag{66}$$

For fixed state and driver, the Z_j are independent standard normals. The generating identity $\mathbb{E}[e^{tZ - t^2/2} e^{uZ - u^2/2}] = e^{tu}$ gives $\mathbb{E} H_a(Z) H_b(Z) = a!\, \mathbf{1}_{a=b}$ by comparing coefficients. Products on distinct coordinate supports are therefore orthogonal. In (64) the singleton H_2 terms from B_j, E_j and C^{mm}_{jj} overlap; the H_1 terms from A_j and C_j also share a degree. The overlapping terms are bounded using $(x+y)^2 \le 2x^2 + 2y^2$. The remaining singleton terms of degrees three and four, and the terms on distinct pair supports, are orthogonal. Consequently

$$\begin{aligned} \int k\mathcal{S}^2 \, dx \le\ & C \sum_j \left(\frac{\mathsf{A}_j^2}{d_j} + \frac{\mathsf{B}_j^2}{d_j^2} \right) + Cs^2 \sum_j \left(\frac{\mathsf{C}_j^2}{d_j} + \frac{\mathsf{E}_j^2}{d_j^2} \right) \\ &+ Cs^2 \sum_{j,k} \left[\frac{(\mathsf{C}^{mm}_{jk})^2}{d_j d_k} + \frac{(\mathsf{C}^{md}_{jk})^2}{d_j d_k^2} + \frac{(\mathsf{C}^{dm}_{jk})^2}{d_j^2 d_k} + \frac{(\mathsf{C}^{dd}_{jk})^2}{d_j^2 d_k^2} \right]. \end{aligned} \tag{67}$$

The constant is independent of the emission count: each singleton or pair support has only a fixed number of terms. Grouping by these supports avoids a crude bound of order n^2 from squaring the sum without using orthogonality.

7.3. **Mesh-uniform chronological bounds.** Put

$$a_{jk} = \sup_{t\in I_j, u\in I_k} |\dot{R}_{i(j),i(k)}(t,u)|.$$

For every fixed $p \geq 2$ the coefficients satisfy, uniformly in the common positive left quadrature,

$$\begin{aligned} \|\mathsf{A}_j\|_p + \|\mathsf{B}_j\|_p &\leq C_p\sqrt{s}e, \\ \|\mathsf{C}_j\|_p + \|\mathsf{E}_j\|_p &\leq C_p a_{jj}, \\ \|\mathsf{C}_{jk}^{mm}\|_p + \|\mathsf{C}_{jk}^{md}\|_p + \|\mathsf{C}_{jk}^{dm}\|_p + \|\mathsf{C}_{jk}^{dd}\|_p &\leq C_p a_{jk}. \end{aligned} \tag{68}$$

Expectations in these norms include both the common driver and the residual state. We prove each bound.

For A_j, $\dot{m}_u e^{y_u}$ is measurable before its Brownian increment. BDG for an adapted step integrand, followed by Minkowski and Hölder, gives

$$\left\| \sum_{u\in j} \dot{m}_u e^{y_u} z_u \right\|_p \leq C_p \left(\sum_{u\in j} w_u \|\dot{m}_u e^{y_u}\|_p^2 \right)^{1/2} \leq C_p\sqrt{s}e.$$

The last inequality uses (57) at order $2p$, uniform Gaussian exponential moments and $\sum w_u = |I_j| \leq d_+$. For B_j there is only an ordinary weighted sum:

$$\left\| \sum_{u\in j} w_u \dot{m}_u e^{2y_u} \right\|_p \leq \sum_{u\in j} w_u \|\dot{m}_u e^{2y_u}\|_p \leq C_p\sqrt{s}e.$$

The factors τ and r_0 are fixed. The coefficient $\dot{R}_{uu}$ is deterministic and bounded by a_{jj}; the same martingale argument bounds C_j, and the weighted-sum argument bounds E_j.

For C_{jk}^{mm}, ordering the nodes by physical left time divides the sum into two strict-order triangles and equal physical times. In the triangle where u precedes v, the contribution, apart from fixed constants, is the adapted outer stochastic sum

$$\sum_{v\in k} e^{y_v} z_v \left(\sum_{\substack{u\in j \\ t_u<t_v}} \dot{R}_{uv} e^{y_u} z_u \right).$$

For each fixed later v, the inner kernel $\dot{R}_{uv}$ is deterministic and bounded by a_{jk}. BDG at order $2p$ gives its norm at most $C_p a_{jk}\sqrt{d_+}$. It is measurable before z_v. Multiplication by e^{y_v} is bounded in L^p by Hölder and exponential moments. A second BDG application to the outer sum gives $C_p a_{jk}$. Exchanging the two nodes gives the same bound for the reversed triangle. This remains valid across asset coordinates because the common Brownian filtration makes the earlier inner sum adapted to the later coordinate's increment.

At equal physical times both states are measurable before the shared cell increments. For either the same or different Brownian coordinates, $\|z_{i,u}z_{k,u}\|_p \leq C_p w_u$. Hölder bounds the exponential left factors, so the equal-time contribution is at most $C_p a_{jk}\sum w_u \leq C_p a_{jk} d_+$. This includes the same-coordinate quadratic-variation terms and the cross-asset equal-time products.

For C_{jk}^{md}, the stochastic node u can precede or follow the ordinary-time node v. In the first case, the sum is

$$\sum_{v\in k} w_v e^{2y_v} \sum_{\substack{u\in j \\ t_u<t_v}} \dot{R}_{uv} e^{y_u} z_u.$$

BDG bounds the inner sum at order $2p$ by $C_p a_{jk}$; Hölder and Minkowski control the outer time sum. In the second case the inner time sum $\sum_{t_v<t_u} \dot{R}_{uv} w_v e^{2y_v}$ is adapted at u and has L^{2p} norm at most $C_p a_{jk} d_+$; the outer stochastic sum is bounded by BDG. For equal physical times an additional time weight multiplies a Brownian increment, and the bound is $C_p a_{jk}\sum w_u^{3/2} \leq C_p a_{jk} d_+^{3/2}$. The transposed ordering proves the dm bound. Finally two applications of Minkowski and Hölder give the dd bound from the two ordinary-time sums

and $\sum_{u\in j} w_u \sum_{v\in k} w_v \le d_+^2$. This proves every assertion in (68). The chronological ordering accounts for the dependence of the state on W^0.

All inverse powers of the integrated variances are uniformly integrable. Weighted Jensen gives

$$D_j \ge |I_j| \exp\left(\frac{2}{|I_j|}\sum_{u\in j} w_u y_u\right), \qquad \mathbb{E}D_j^{-p} \le d_-^{-p} e^{C_p}. \tag{69}$$

The weighted state average in this display is unconditionally Gaussian with mean bounded by the state-mean envelope and variance at most Cs_0: its weights are positive and sum to one, and all state covariances are bounded by Cs_0. Taking its exponential moment proves the negative-moment bound for every fixed p.

Hölder's inequality applied to (67), with (68) at sufficiently high fixed moment orders and (69), with $d_j = r_0 D_j$, bounds the information as follows. The state-mean terms contribute $Cmse^2$. Covariance singletons and pairs contribute $Cs^2 \sum_{j,k} a_{jk}^2$. For example $\mathbb{E}(\mathsf{C}_{jk}^{mm})^2/(d_j d_k) \le Ca_{jk}^2$ follows by a fourth coefficient moment and a higher inverse moment; the other denominator powers are handled identically. Constants may depend on $r_0 > 0$. By (56),

$$\sum_{j,k} a_{jk}^2 \le Cm \le Cn. \tag{70}$$

Indeed $(1+x)e^{-cx} \le C_c e^{-cx/2}$, and physical intervals separated by l indices have distance at least $(l-1)d_-$ once $l \ge 2$. There are only two asset copies per interval, so for each j the sum in k is bounded by a geometric series; adjacent and coincident intervals contribute finitely many bounded terms. Equations (66)–(70) prove the finite-quadrature bound

$$I_\xi \le Cn(s^2 + se^2), \tag{71}$$

independent of refinement, emission count and the true small parameters.

7.4. **Exact increments and Hellinger contraction.** We justify both the parameter integration and the passage to the exact stochastic-integral experiment. For a fixed finite quadrature, the conditional node mean has the C^1 version established above, and the conditional Gaussian covariance is positive definite. Differentiation under the finite Gaussian integral, first with localization, gives (64). On a compact parameter interval the higher moments in (68) and (69) permit removing localization by Hölder and dominated integral estimates. The resulting positive marginal density is pointwise absolutely continuous in ξ for almost every driver and return vector. The integrable bounds on the localized derivative allow the fundamental theorem of calculus to pass through the finite mixture by Fubini. The driver law itself does not change with ξ.

Relative to that law times Lebesgue measure for returns, $\partial_\xi \sqrt{p_\xi} = (\partial_\xi p_\xi)/(2\sqrt{p_\xi})$ has squared L^2 norm $I_\xi/4$. The Jensen and moment bounds control this norm uniformly on the parameter interval. Thus pointwise absolute continuity, Fubini and Minkowski justify integrating the square-root derivative in the joint L^2 space, giving

$$\mathsf{H}(p_\xi, p_{\xi'}) \le \frac{1}{2}\int_\xi^{\xi'} \sqrt{I_u}\,|du| \le C\sqrt{n(s^2+se^2)}|\xi - \xi'|. \tag{72}$$

This derives the Hellinger bound from the finite-dimensional score.

For each fixed n, ξ, the positive left sums for D_j converge almost surely to the strictly positive integrals in (58), since the state paths are continuous. The left mean sums converge in L^2 by Itô isometry: the integrands converge pointwise by continuity, and Gaussian exponential moments uniformly dominate their squared differences on the finite horizon. Normal kernels with means converging in probability and variances converging to positive values converge in total variation in probability. The deterministic kernel statement follows from pointwise density convergence and Scheffé's lemma. For the finite product of emissions the same conclusion holds. Its total variation is bounded, so its expectation tends to zero. Coupling the conditional kernels with the same (Y, W^0) therefore gives total-variation convergence of the joint law of the emissions and the additional observed driver path.

Since $\mathsf{H}^2 \le 2\,\mathrm{TV}$, the triangle inequality and this convergence pass the mesh-uniform bound (72) to exact richer observations. Hellinger contracts under a parameter-independent kernel: conditional Cauchy–Schwarz increases the affinity $\int \sqrt{pp'}$ after a kernel, and $\mathsf{H}^2 = 2 - 2\int \sqrt{pp'}$. Applying the reconstruction and noise kernel after (52) gives (51) for every original noisy level. This completes the proof of Lemma 7.1.

7.5. **Two-point risk and honest-diameter lower bounds.** We work on the drift ball in Proposition 3.2, with $V = I$, $q = 0$, and fixed admissible b, μ, ν, s and signed L. For $0 \le e \le \sqrt{1-\kappa}$ use $L = eE_{12}$; for the whole range $0 \le e \le \sqrt{2(1-\kappa)}$ use $L = (e/\sqrt{2})I$. Fix any Frobenius-unit direction D and a short line $M_\xi = M_0 + \xi D$ inside the full four-dimensional ball. Its symmetric part remains uniformly negative, so $A_{0,\xi} = \mathrm{chol}[-(M_\xi + M_\xi^\top)]$ is C^2 with bounded derivatives. Class, strip and excitation conditions hold uniformly along the line. At $e = 0$ the excitation condition holds by its zero convention. The nuisance restrictions are used solely to construct the lower submodel, not to inform the estimator.

In $\Delta = 1$ units choose two interior points separated by

$$d = c\min\{1, [n(s^2 + se^2)]^{-1/2}\}, \tag{73}$$

with fixed c small enough for both to stay in the ball and for Lemma 7.1 to give $\mathrm{TV} \le 1/2$. Any estimate of two values a distance d apart defines a test by selecting the nearer value. On a classification error its squared loss is at least $d^2/4$. The sum of error probabilities of any test is at least $1 - \mathrm{TV}$, so the maximum squared risk is at least $d^2(1 - \mathrm{TV})/8$. This classical two-point argument applies to every measurable function of all observed levels and proves the matrix lower bound in (16). Restoring fixed Δ units changes only constants. For a nonzero contrast choose $D = C/\|C\|_F$ in dimensionless drift units. Its contrast separation is $\|C\|_F d$, proving the scalar order including either off-diagonal entry. The upper order follows from (14) and Cauchy–Schwarz. This proves Theorem 3.1(C), including the rank-one lower subfamily.

Let a confidence set be honest at these two points, and let their contrast values be ψ_0, ψ_1. Under the first law,

$$\mathbb{P}_0\{\psi_0, \psi_1 \text{ both in the set}\} \ge 1 - 2\alpha - \mathrm{TV}(P_0, P_1) \ge 1/2 - 2\alpha. \tag{74}$$

Coverage of ψ_1 transfers from P_1 to P_0 with loss at most TV; intersecting with coverage of ψ_0 proves the inequality. For fixed $\alpha < 1/4$, the expected squared diameter is consequently at least $(1/2 - 2\alpha)|\psi_1 - \psi_0|^2$. This proves the matching worst-case honest-diameter lower bound in Theorem 3.1(D). At $s = 0$, two distinct drifts with identical regular nuisance parameters have the same law. The same event has probability at least $1 - 2\alpha$, so honest structural uncertainty cannot uniformly shrink. Together with the previously proved upper statements, this finishes Theorem 3.1.

8. Absent and nonexciting leverage

The same price experiment gives two contrasts in how leverage affects learnability. With small unknown leverage, including zero, the covariance-only rate remains attainable and has a matching worst-case bound on a specified subfamily. In a diagonal state model with known cross-loading support, the leverage term improves the rate for one persistence while the other retains the covariance-only rate.

8.1. **Small unknown full leverage: a worst-case covariance boundary.** Let $\mathcal{G}(s_0, \ell_0)$ contain all parameters of the original compact model with $\|L\|_F \le \ell_0$ and a fixed positive residual margin, allowing $q = 0, L = 0$. Its loading support and orientation are unknown, and it has no excitation condition.

Proposition 8.1. *For sufficiently small fixed $\ell_0, s_0 > 0$, all positive-scale parameters in $\mathcal{G}(s_0, \ell_0)$ are identified, zero and positive scales are separated, and a single full-class exact-moment estimator has squared dimensionless drift risk at most $C\min\{1, (ns^2)^{-1}\}$. If the drift/shape class*

contains

$$M_\gamma = \begin{pmatrix} -a_1 & \gamma \\ \gamma & -a_2 \end{pmatrix}, \quad V = I, \quad a_1 \neq a_2 > 0, \quad \gamma \in J \Subset (0, \sqrt{a_1 a_2}), \tag{75}$$

where J has positive length, this is the matching worst-case order at fixed s, also for the off-diagonal contrast on that subfamily.

Proof. We again work in $\Delta = 1$ units. The first-cubic profile of Lemma 5.2 remains valid without excitation. On this class $h_* \le C\sqrt{s_*}\ell_0$. Lemma 5.1, applied to $G^0 = G - J$, and (33) give

$$\rho_0 := \delta_\Sigma + s_*\delta_M \le C\epsilon + Ch_*\epsilon + Ch_*^2(\delta_M + \delta_\Sigma) \le C\epsilon + C\ell_0^2(1 + s_0)\rho_0.$$

After fixing s_0 for the cubic profile and covariance inverse, we choose ℓ_0 so that the last coefficient is at most $1/2$. Absorbing the last term gives $\rho_0 \le C\epsilon$. The same regular formulas as before recover b, ν, λ, μ, q. At positive scale, equality gives M, Σ, then H from the first cubic and $L = A^{-1}H$. At zero scale only the regular parameters remain, and the bound on δ_Σ separates it from positive scale.

The full-class raw variance lemma and dense minimum-distance construction, now over $\mathcal{G}$, give a uniform C/n squared bound for ρ_0 plus regular nuisance errors. The truth drift weight is at least s; division and compact capping give the upper rate. The extra cubic lag in our 27 features is harmless; the inverse only needs the covariance pair and first cubic.

For the lower bound, we use (75), $L = 0, Q = I$ and fixed admissible nuisance parameters and s. This symmetric drift submodel contains an interval of positive length. Lemma 7.1 at $e = 0$ gives $\mathsf{H}^2 \le Cns^2|\gamma - \gamma'|^2$; its proof here is covariance-only since the conditional state mean is fixed. Choosing an interior pair of separation $c\min\{1, (\sqrt{n}s)^{-1}\}$ and applying the all-price two-point test proves the lower order, also for $M_{12} = \gamma$. □

This worst-case result uses the admitted zero-leverage subfamily. It is not a pointwise impossibility claim at every small nonzero loading. Dynamic excitation in Theorem 3.1 gives another inverse route, without a small-leverage condition on its performance subset.

8.2. **A known-support nonexciting family.** We next consider the known restrictions

$$\begin{gathered} M = \operatorname{diag}(-a_1, -a_2), \qquad \Sigma = \operatorname{diag}(v_1, v_2), \qquad v_i = sV_i, \quad V_1 + V_2 = 2, \\ A = \operatorname{diag}(\sqrt{2a_1 v_1}, \sqrt{2a_2 v_2}), \qquad L = \ell E_{12}. \end{gathered} \tag{76}$$

The a_i range independently over fixed positive compact intervals of positive length. The V_i range over compact positive shapes with $(1, 1)$ admitted. Signed ℓ, scale, shape, q, μ, b, ν are all unknown, with the previous nuisance bounds and

$$q^2 + \ell^2 \le 1 - \kappa. \tag{77}$$

This is a separate compact class $\mathcal{D}(s_0)$: its diagonal state structure and leverage support are known, not an arbitrary unknown rank-one restriction. The determinant and trace of $Q - L^\top L = \left(\begin{smallmatrix} 1 & q \\ q & 1-\ell^2 \end{smallmatrix}\right)$ show that its least eigenvalue is at least $\kappa/2$, so all required uniform residual bounds hold with that fixed margin. At $\ell \neq 0$, H and ZH have only their first rows nonzero. Thus their concatenation has row rank one and does not satisfy (10).

Theorem 8.2. *For a sufficiently small fixed s_0, all positive-scale parameters in $\mathcal{D}(s_0)$ are exactly identified. One exact-moment estimator over the whole class, with all parameters unknown, attains simultaneously*

$$\begin{gathered} \mathbb{E}(\widehat{a}_1 - a_1)^2 \le Cr_1, \qquad \mathbb{E}(\widehat{a}_2 - a_2)^2 \le Cr_2, \\ r_1 = \min\{1, [n(s^2 + s\ell^2)]^{-1}\}, \quad r_2 = \min\{1, (ns^2)^{-1}\}. \end{gathered} \tag{78}$$

On each slice $s > 0$, $|\ell| = e \le \sqrt{1 - \kappa}$, these are matching minimax orders for every estimator using all noisy levels. For each fixed $c = (c_1, c_2)$ the contrast $c_1 a_1 + c_2 a_2$ has order $c_1^2 r_1 + c_2^2 r_2$. Full-class honest projected confidence sets have this expected squared diameter upper order for every fixed $\alpha \in (0, 1)$, and matching worst-case lower order for fixed $\alpha < 1/4$.

We prove the upper inverse first and then the two separate all-price obstructions; the latter cannot be inferred from the moment derivatives.

8.3. **Exact directional inverse and upper bounds.** In (76), $K_{12}(t) = \sqrt{2a_1v_1}\ell e^{-a_1t}$, $C_{12}(t) = 0$, and $K_{11} = K_{22} = 0$. Lemma 4.1 gives, exactly,

$$T_{k,12} = 2h^2\lambda_1\sqrt{\lambda_2}e^{-v_2/2}\sqrt{2a_1v_1}\ell f_h(a_1)e^{-a_1k\Delta}, \qquad f_h(a) = \frac{2(\cosh(ah)-1)}{a^2h^2} > 0. \tag{79}$$

The factor is $f_h(a) = A_h(-a)A_h(a)$ by direct integration. Both own-coordinate corrections $J_{k,ii}$ vanish, since their first loading factor K_{ii} is zero. The own fourth-moment pairs are therefore exactly their scalar covariance-only maps, for every q, ℓ.

The scalar proof of Lemma 5.1 gives, for any two candidates in this entire restricted class,

$$|v_i - v_i'| + \max(v_i, v_i')|a_i - a_i'| \le C\|F(\theta) - F(\theta')\|, \qquad i = 1, 2. \tag{80}$$

Specifically the scalar linear pair is $B_1 = 4v_i f_h(a_i)e^{-a_i\Delta}$, $B_2 = e^{-a_i\Delta}B_1$; the larger variance endpoint has $|B_1'| \ge c\max(v_i, v_i')$. The scalar logarithm of B_2/B_1 controls persistence on the positive compact interval. Covariance and drift remainder derivatives are $O(v_*)$ and $O(v_*^2)$, as before, and are absorbed for small s_0.

Put $z(a) = e^{-a\Delta}$. Equation (79) gives $T_{2,12} = z(a_1)T_{1,12}$ and, for any alternative,

$$|T_{1,12}|\,|z(a_1) - z(a_1')| \le |T_{2,12} - T_{2,12}'| + |z(a_1')|\,|T_{1,12} - T_{1,12}'|.$$

On the compact interval $|z(a) - z(a')| \ge c|a - a'|$, and the exact amplitude obeys $|T_{1,12}| \ge c\sqrt{v_1}|\ell|$. The centered transformations are Lipschitz in the raw features. Thus, without knowing magnitude or sign at either endpoint,

$$\sqrt{v_1}|\ell|\,|a_1 - a_1'| \le C\|F(\theta) - F(\theta')\|. \tag{81}$$

Together with (80), this gives the truth-side bounds

$$\sqrt{v_1^2 + v_1\ell^2}\,|a_1 - a_1'| \le C\|F - F'\|, \qquad v_2|a_2 - a_2'| \le C\|F - F'\|. \tag{82}$$

They include comparison with zero-scale alternatives. Since v_i is comparable to s, Lemma 6.1 and the measurable full-class minimum-distance construction prove both upper rates in (78). In this statement full class means the entire class $\mathcal{D}(s_0)$, not the unrestricted class with unknown loading support. One estimator attains both rates simultaneously.

At positive scale (80) identifies a_i, v_i, including at $\ell = 0$. Equation (79) then identifies the signed ℓ, and the regular formulas identify b, μ, ν, q; s, V_i follow. The scalar covariance inverse also separates zero scale. For a contrast estimate, $(c_1x + c_2y)^2 \le 2c_1^2x^2 + 2c_2^2y^2$ gives upper order $c_1^2r_1 + c_2^2r_2$.

The confidence set (49), constructed over $\mathcal{D}(s_0)$, has uniform coverage by the same moment MSE, including at zero scale. Every candidate has moment error at most Z_n about the truth off or on the coverage event. Equation (82) bounds the coordinate diameters separately, with each capped by its compact interval. The expected squared contrast diameter is at most $C_\alpha(c_1^2r_1 + c_2^2r_2)$. This derives the directional confidence upper claim without a plug-in leverage estimate.

8.4. **All-price lower bound for the exposed persistence.** For a_1, we restrict to $q = 0$, $v_1 = v_2 = s$, fixed signed ℓ, fixed a_2 and fixed admissible regular nuisance parameters. With $\varrho = 1 - \ell^2$, we use the representation

$$B_1 = \ell U_2 + \sqrt{\varrho}W_1$$

with U_1, U_2, W_1, B_2 independent standard Brownian motions. The enlarged experiment observes the entire prehistory of U_2 and the Y_2 path, together with every clean asset-1 increment of the full grid. The law of the two paths is independent of $a = a_1$. The asset-2 clean levels can be reconstructed from U_2, Y_2 with their fixed starting level; adding the original level errors to both assets is a parameter-independent kernel. The target Y_1 stays hidden.

Given U_2, its conditional mean and covariance are

$$m_a(t) = \mu_1 + \ell\sqrt{2as}\int_{-\infty}^{t} e^{-a(t-u)}dU_2(u), \qquad s\varrho e^{-a|t-u|}. \tag{83}$$

Since U_1 is independent of this state, the emissions conditional on Y_1 have fixed means $b_1|I_j|$ and variances $D_j = \int_{I_j} e^{2Y_1(t)}dt$.

For finite quadratures put

$$J_j^{\mathrm{m}} = \sum_{u\in j} w_u \dot m_a(t_u)e^{2y_u}, \qquad W_{jk} = \sum_{u\in j, v\in k} w_u w_v \dot R_a(t_u,t_v)e^{2y_u+2y_v}, \quad R_a(t,u) = e^{-a|t-u|}.$$

With $S_j = H_2(Z_j)/(2D_j)$ and $T_j^{\mathrm{H}} = H_4(Z_j)/(4D_j^2)$, define

$$\mathcal{Z} = \sum_j W_{jj}T_j^{\mathrm{H}} + 2\sum_{j<k} W_{jk}S_jS_k.$$

Equation (61) gives the complete derivative

$$\partial_a p = \mathbb{E}\left[k\left\{2\sum_j J_j^{\mathrm{m}}S_j + 2s\varrho\mathcal{Z}\right\} \mid U_2\right]. \tag{84}$$

Here $\dot R_a(t,t) = 0$, so there is no covariance-diagonal derivative term in this scalar fixed-variance submodel. The state-mean derivative term is retained. Under k, its singleton H_2 supports are orthogonal to the covariance singleton H_4 and pair H_2H_2 supports. Since $|W_{jk}| \le a_{jk}D_jD_k$, where $a_{jk} = \sup_{I_j,I_k}|t-u|e^{-a|t-u|}$, Jensen and Hermite norms give

$$I(a) \le 2\sum_j \mathbb{E}[(J_j^{\mathrm{m}})^2/D_j^2] + Cs^2\varrho^2\sum_{j,k} a_{jk}^2. \tag{85}$$

For instance $\mathbb{E}_k(T_j^{\mathrm{H}})^2 = 3/(2D_j^4)$ and $\mathbb{E}_k S_j^2 = 1/(2D_j^2)$, which cancel the D factors in the covariance part.

Differentiating the mean convolution gives

$$\dot m_a(t) = \ell\sqrt{2as}\int_{-\infty}^{t}[1/(2a) - (t-u)]e^{-a(t-u)}dU_2(u), \qquad \mathbb{E}\dot m_a(t)^2 = \frac{s\ell^2}{4a^2}. \tag{86}$$

The variance identity follows by integrating $2as\ell^2\int_0^\infty(1/(2a)-v)^2e^{-2av}dv$. For the bounded-interval supremum required to control J_j^{m}, let $x = m_a - \mu_1$. It solves the stable two-equation system

$$dx = -ax\,dt + \ell\sqrt{2as}\,dU_2, \qquad d\dot m_a = -(x + a\dot m_a)dt + \ell\sqrt{s/(2a)}\,dU_2.$$

Its stationary starting second moments are $\mathbb{E}x^2 = s\ell^2$ and those in (86). On an interval of length at most d_+, the integrated equations, Cauchy–Schwarz for drift integrals and Doob's L^2 inequality for the Brownian terms give an inequality of the form

$$\mathbb{E}\sup_{t_0\le t\le t_0+u}(|x(t)|^2 + |\dot m_a(t)|^2) \le Cs\ell^2 + C\int_0^u \mathbb{E}\sup_{t_0\le t\le t_0+v}(|x(t)|^2 + |\dot m_a(t)|^2)dv.$$

Gronwall yields $\mathbb{E}\sup_{I_j}|\dot m_a|^2 \le Cs\ell^2$, uniformly on the positive compact a interval. Positive weights imply $|J_j^{\mathrm{m}}|/D_j \le \sup_{I_j}|\dot m_a|$. The same geometric summability as (70) bounds $\sum a_{jk}^2$ by Cn. Thus (85) gives $I(a) \le Cn(s^2 + s\ell^2)$ uniformly in quadrature. The parameter integration and exact-kernel convergence proved after (71) apply here, including the additional observed paths. After reconstruction,

$$\mathsf{H}^2(P_a^{(n)}, P_{a'}^{(n)}) \le Cn(s^2 + s\ell^2)|a-a'|^2. \tag{87}$$

An interior pair with separation $c\sqrt{r_1}$ therefore proves the all-price r_1 lower order.

8.5. **All-price lower bound for the unexposed persistence.** We now fix a_1 and vary $a = a_2$, again with $q = 0$, $v_i = s$, fixed signed ℓ and nuisance parameters. Here the enlarged experiment observes the parameter-free B_1, U_1 on their full prehistory, together with all clean asset-2 increments. These observations reconstruct asset-1 prices using the fixed a_1. We use the representation

$$U_2 = \ell B_1 + \sqrt{\varrho} W_2$$

with B_1, W_2, B_2, U_1 independent. The hidden Y_2 is independent of B_1, and the exact conditional emission means and variances are

$$a_j = b_2|I_j| + \ell \int_{I_j} e^{Y_2(t)} dB_1(t), \qquad d_j = \varrho D_j, \quad D_j = \int_{I_j} e^{2Y_2(t)} dt. \tag{88}$$

The state dependence of the emission means requires retaining their derivatives.

In positive left quadratures, the B_1 increments z_u are independent, with variance w_u. The emission derivatives are now

$$\mathfrak{m}_u = \ell e^{y_u} z_u, \qquad \mathfrak{d}_u = 2\varrho w_u e^{2y_u}.$$

The state mean is fixed and its covariance is sR_a, $R_a(t,u) = e^{-a|t-u|}$. Only in this scalar fixed-variance model is $\dot{R}_a(t,t) = 0$. All four arrays $\mathsf{C}^{mm}, \mathsf{C}^{md}, \mathsf{C}^{dm}, \mathsf{C}^{dd}$ from (63) enter with these derivatives. The exact marginal score derivative is

$$\partial_a p = (s/2)\mathbb{E}[k\mathcal{Q} \mid B_1], \tag{89}$$

where $\mathcal{Q}$ is the full expression (65); there are no mean-state or covariance-diagonal additions in this particular submodel. All singleton degrees and pair supports in $\mathcal{Q}$ are orthogonal here. The Hermite identities give the exact norm

$$\begin{aligned} \int k\mathcal{Q}^2\, dx = \sum_j \left[\frac{2(\mathsf{C}^{mm}_{jj})^2}{d_j^2} + \frac{6(\mathsf{C}^{md}_{jj})^2}{d_j^3} + \frac{3(\mathsf{C}^{dd}_{jj})^2}{2d_j^4} \right] \\ + 4\sum_{j<k} \left[\frac{(\mathsf{C}^{mm}_{jk})^2}{d_j d_k} + \frac{(\mathsf{C}^{md}_{jk})^2}{2d_j d_k^2} + \frac{(\mathsf{C}^{dm}_{jk})^2}{2d_j^2 d_k} + \frac{(\mathsf{C}^{dd}_{jk})^2}{4d_j^2 d_k^2} \right]. \end{aligned} \tag{90}$$

Here the mm, md and dm terms account for the random mean, and the dd terms for the variance mixture.

The integrated-variance denominators cancel as follows. Let

$$a_{jk} = \sup_{I_j, I_k} |\dot{R}_a(t,u)|.$$

Conditional on the full Y_2 path, the z_u are independent centered Gaussians, since Y_2 is independent of B_1 in this richer experiment. Their second and fourth moments give

$$\begin{aligned} \mathbb{E}_{B_1}(\mathsf{C}^{mm}_{jj})^2 &\le 2\ell^4 a_{jj}^2 D_j^2, \\ \mathbb{E}_{B_1}(\mathsf{C}^{mm}_{jk})^2 &\le \ell^4 a_{jk}^2 D_j D_k \quad (j \ne k), \\ \mathbb{E}_{B_1}(\mathsf{C}^{md}_{jk})^2 &\le 4\ell^2 \varrho^2 a_{jk}^2 D_j D_k^2, \\ |\mathsf{C}^{dd}_{jk}| &\le 4\varrho^2 a_{jk} D_j D_k. \end{aligned} \tag{91}$$

The transposed dm inequality interchanges j, k. To see the first bound, C^{mm}_{jj} is a Gaussian quadratic form with zero conditional mean, since its diagonal coefficients $\dot{R}_a(t_u, t_u)$ vanish. Its variance is twice the weighted sum of squared off-diagonal coefficients:

$$2\ell^4 \sum_{u,v \in j} \dot{R}_a(t_u, t_v)^2 e^{2y_u + 2y_v} w_u w_v \le 2\ell^4 a_{jj}^2 D_j^2.$$

For distinct emissions the two independent Brownian sums give the second bound without the factor two. For md, conditional variance of the linear Gaussian sum is

$$\ell^2 \sum_{u \in j} w_u e^{2y_u} \left(2\varrho \sum_{v \in k} \dot{R}_a(t_u, t_v) w_v e^{2y_v} \right)^2 \le 4\ell^2 \varrho^2 a_{jk}^2 D_j D_k^2.$$

The *dd* bound is the absolute bound for two deterministic weighted sums. These bounds hold uniformly in the number of quadrature nodes within each emission.

After substitution of $d_j = \varrho D_j$ into (90), the three singleton contributions are bounded respectively by $4\ell^4/\varrho^2$, $24\ell^2/\varrho$, and 24, times a_{jj}^2. The analogous pair bounds are controlled by the same fixed envelope. Consequently

$$\mathbb{E}\int k\mathcal{Q}^2\,dx \le 24\left(1+\frac{\ell^2}{\varrho}+\frac{\ell^4}{\varrho^2}\right)\sum_{j,k}a_{jk}^2 \le \frac{24}{\varrho^2}\sum_{j,k}a_{jk}^2. \tag{92}$$

The last inequality follows from $\varrho^2+\ell^2\varrho+\ell^4\le 1$ with $\varrho = 1-\ell^2$. There are no negative powers of D_j left. Jensen for (89), summability and $\varrho \ge \kappa$ give

$$I(a) \le Cns^2/(1-\ell^2)^2 \le C_\kappa ns^2. \tag{93}$$

Finite quadratures, parameter integration and convergence to exact conditional kernels are justified exactly as in the proof of Lemma 7.1. Here the target state is independent of the additional observed driver, but the emission mean remains stochastic. Reconstruction of asset-1 prices and addition of one error at each distinct time to both assets give

$$\mathsf{H}^2(P_a^{(n)}, P_{a'}^{(n)}) \le C_\kappa ns^2|a-a'|^2. \tag{94}$$

An interior a_2 pair with separation $c\sqrt{r_2}$ proves its all-price lower order. In particular, fixed nonzero cross leverage does not accelerate this persistence.

For $c_1a_1+c_2a_2$, each coordinate can vary independently over an interval of positive length. The two testing arguments lower-bound worst-case risk by constants times $c_1^2r_1$ and $c_2^2r_2$; their maximum is at least half their sum. The upper bound was proved by the same coordinatewise argument after (82). For honest sets, (74) applies separately to these same two-point submodels. The maximum-to-sum inequality then gives, for fixed $\alpha < 1/4$, the matching expected squared diameter lower order. All infima here permit every estimator using the full noisy grid; fixing nuisance parameters on a lower submodel does not restrict the procedure. This completes Theorem 8.2. None of these rate bounds is extended to perfect leverage; the constant in (93) explicitly uses its positive residual margin.

9. Conclusion

The precision warranted by noisy prices depends on where leverage carries information through the hidden dynamics. In the coupled rank-one example, three exact lags identify all four drift entries. In the separate diagonal known-support example, the second persistence remains limited by covariance-only information. The rates quantify this difference even when all nuisance parameters are unknown. Their lower bounds use every noisy price level, and honest confidence sets express the resulting uncertainty without assuming excitation for coverage. The precision guarantees retain fixed excitation and residual margins and a sufficiently small, unevaluated covariance ceiling; they do not yet give calibrated sample sizes or practical confidence widths.

Acknowledgements

The author used ChatGPT (OpenAI, GPT-6 Astra) to assist in drafting and polishing the exposition from the author's notes and proof sketches, as well as in checking for possible gaps and inconsistencies. The author is responsible for the final statements and proofs.

References

[1] X. Cheng, E. Renault and P. Sangrey, *Identifying the Volatility Risk Price Through the Leverage Effect*, author manuscript, 23 April 2024. Publication DOI: 10.1016/j.jeconom.2024.105943.

[2] Md. N. Ahsan and J.-M. Dufour, *High-frequency instruments and identification-robust inference for stochastic volatility models*, October 2024 author manuscript, including Supplementary Appendix. Journal publication (2025) DOI: 10.1111/jtsa.12812.

[3] Md. N. Ahsan, J.-M. Dufour and G. Rodriguez-Rondon, *Estimation and Inference for Higher-Order Stochastic Volatility Models With Leverage*, Journal of Time Series Analysis **46**(6) (2025), 1064–1084. DOI: 10.1111/jtsa.12851.

[4] Stéphane Gregoir and N. Meddahi, *Towards Identification of Shocks in Linear State-Space Models: Application to Stochastic Volatility Model*, Journal of Time Series Analysis **46**(6) (2025), 1205–1223; first published 8 September 2025. DOI: 10.1111/jtsa.70012.

[5] S. Oymak and N. Ozay, *Non-asymptotic Identification of LTI Systems from a Single Trajectory*, arXiv:1806.05722v2, 3 February 2019.

[6] J. Masoliver and J. Perelló, *Multiple time scales and the exponential Ornstein–Uhlenbeck stochastic volatility model*, arXiv:cond-mat/0501639v1, 26 January 2005; Quantitative Finance **6** (2006), 423–433. DOI: 10.1080/14697680600727547.

[7] A. Gloter, *Efficient estimation of drift parameters in stochastic volatility models*, Finance and Stochastics **11** (2007), 495–519. DOI: 10.1007/s00780-007-0048-2.

[8] K. Abraham, Z. Naulet and É. Gassiat, *Fundamental limits for learning hidden Markov model parameters*, arXiv:2106.12936v3, 24 October 2022.

[9] C. H. Chong, M. Hoffmann, Y. Liu, M. Rosenbaum and G. Szymanski, *Statistical inference for rough volatility: Minimax theory*, arXiv:2210.01214v2, 15 February 2024.